

- Q.23 What is the purpose of Flight Testing?
 Q.24 Explain the term endurance and range?
 Q.25 Describe Semi-monocoque type of construction and its merits.
 Q.26 Discuss various jet engines.
 Q.27 What are the basic instruments used in Avionics system?
 Q.28 How does an airplane fly?
 Q.29 Explain the working of Air speed Indicator.
 Q.30 How the tail contributes to the stability of the airplane.
 Q.31 Draw the stress strain diagram for steel.
 Q.32 What type of loads act on airplane wing?
 Q.33 Name the aircraft instruments with their application.
 Q.34 Briefly describe about the various types of drag acting on aircraft
 Q.35 What is the use of hydraulic system in aircraft?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain aircraft fuel system. Why the fuel is stored in the wings?
 Q.37 Explain Aircraft Stability in detail with the contribution of different parts in it.
 Q.38 Explain the different types of materials used in Aircraft manufacturing ? Why all metal aircraft is better?

(20)

(4)

187733/147733

No. of Printed Pages : 4
 Roll No.

187733/147733

3rd Sem / Aircraft Maintenance Engg. Subject:- Introduction to Aeronautics

Time : 3Hrs.

M.M. : 100

SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (10x1=10)
 Q.1 What type of aircraft is primarily designed for transporting passengers over long distances?
 a) Fighter jet b) Cargo plane
 c) Commercial airliner d) Helicopter
 Q.2 What is the main purpose of emergency exits in an aircraft?
 a) To allow passengers to board the aircraft
 b) To provide additional ventilation during flight
 c) To facilitate rapid evacuation in case of an emergency
 d) To allow crew members to enter and exit the aircraft
 Q.3 What is endurance?
 a) Maximum time of flight of aircraft
 b) Maximum load it can carry
 c) Maximum take off weight
 d) None of the above
 Q.4 What type of construction method utilizes materials such as aluminium or titanium to create robust and reliable structures?

(1)

187733/147733

- a) Composite construction
 - b) Stressed construction
 - c) Monocoque construction
 - d) Metal construction
- Q.5 Which of the following is an advantage of using hydraulic systems in industrial applications?
- a) High initial cost
 - b) Limited power transmission capabilities
 - c) High power-to-weight ratio
 - d) High risk of fluid leakage
- Q.6 What is the primary purpose of contamination control filters in hydraulic systems?
- a) To regulate fluid flow rate
 - b) To prevent fluid leakage
 - c) To remove impurities and particles from the fluid
 - d) To control the direction of fluid flow
- Q.7 Which type of airframe construction uses tubular structures?
- a) Monocoque b) Braced Monocoque
 - c) Semimonocoque d) Truss
- Q.8 Which of the following statements about jet propulsion is true?
- a) Jet engines operate on the principle of buoyancy
 - b) Jet propulsion is only suitable for aircraft
 - c) Jet engines generate thrust by expelling high-velocity exhaust gases
 - d) Jet engines rely on external propellers for propulsion

- Q.9 Which engine has the highest efficiency?
- a) Turbo Jet Engine b) Turbo Fan Engine
 - c) Rocket Engine d) Turbo Shaft Engine
- Q.10 In a turbine engine, what does a bleed valve do?
- a) Regulates the airflow into the combustion chamber
 - b) Controls the temperature of the engine's air supply
 - c) Releases excess air from the compressor section
 - d) Seals off the air inlet during engine shutdown

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What type of aircraft did the Wright brothers build?
- Q.12 The maximum improvement happened in which parameter of the flight over 100 years?
- Q.13 Write the equation of the lift.
- Q.14 How does a Turbojet differ from a Turbo Fan engine?
- Q.15 What is L/D ratio?
- Q.16 What is STOL?
- Q.17 Name two secondary control surfaces.
- Q.18 Which thermodynamic cycle is used in a gas turbine?
- Q.19 Where are composites used in an aircraft?
- Q.20 What is the input in ROC indicator?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Draw drag curve and explain drag bucket?
- Q.22 Explain the working of an altimeter?